Workshop: Concentrations

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Conversion between concentrations

1. Convert the following concentrations into mg/ml
   a. 100 mg in 5 ml

$$\frac{100}{5} = 20 \text{ mg/ml}$$

   b. 1000 mg in 20 ml

$$\frac{1000}{20} = 50 \text{ mg/ml}$$

   c. 1 g in 100 ml

$$\frac{1000}{100} = 10 \text{ mg/ml}$$

   d. 1 g in 200 ml

$$\frac{1000}{200} = 5 \text{ mg/ml}$$

   e. 2 g in 500 ml

$$\frac{2000}{500} = 4 \text{ mg/ml}$$

   f. 10 g in 1 l

$$\frac{10000}{1000} = 10 \text{ mg/ml}$$

   Strongest = b, weakest = e

2. Convert the following concentrations into %w/v. %w/v means g/100 ml.

$$\%\text{w/v} = \frac{\text{mass (g)}}{\text{volume (ml)}} \times 100$$

   a. 1 g in 100 ml

$$\frac{1}{100} \times 100 = 1\% \; w/v$$

   b. 5 g in 100 ml

$$\frac{5}{100} \times 100 = 5\%\, w/v$$

c. 5 g in 1000 ml

$$\frac{5}{1000} \times 100 = \frac{5}{10} = 0.5\ \%\, w/v$$

d. 5 g in 50 ml

$$\frac{5}{50} \times 100 = 5 \times 2 = 10\%\, w/v$$

e. 500 mg in 10 ml

$$\frac{0.5}{10} \times 100 = 0.5 \times 10 = 5\%\, w/v$$

f. 150 mg/ml

$$\frac{0.15}{1} \times 100 = 15\%\, w/v$$

B and E have the same strength.

3. Convert each of the following concentrations into the given unit.
   a. 250 mg/10 ml into mg/ml

$$\frac{250}{10} = 25\ mg/ml$$

   b. 250 mg/10 ml into %w/v

$$\frac{0.25}{10} \times 100 = 2.5\ \%\, w/v$$

   c. 40% w/v into mg/ml

$$40\ \% \frac{w}{v} = \frac{40g}{100ml} = \frac{40000\ mg}{100\ ml} = 400\ mg/ml$$

   d. 2% w/v into mg/ml

$$2\ \% \frac{w}{v} = \frac{2g}{100ml} = \frac{2000\ mg}{100\ ml} = 20\ mg/ml$$

4. Convert each of the following concentrations into %v/v. %v/v can be thought of as ml/100 ml (or l/100 l).
   a. 50 ml in 200 ml

$$\frac{50}{200} = \frac{25}{100} = 25\ \%\ v/v$$

b. 2 ml in 50 ml

$$\frac{2}{50} = \frac{4}{100} = 4\ \%\ v/v$$

c. 500 ml in 2 l

$$\frac{500}{2000} = \frac{250}{1000} = \frac{25}{100} = 25\ \%\ v/v$$

d. 0.5 ml in 500 ml

$$\frac{0.5}{500} = \frac{0.1}{100} = 0.1\ \%\ v/v$$

Ordering concentrations by strength

5. Order the following concentrations from weakest to strongest.

   A: 1% w/v

$$\frac{1\ g}{100\ ml} = \frac{1000\ mg}{100\ ml} = 10\ mg/ml$$

   B: 300 mcg/ml

$$\frac{0.3}{1} = 0.3\ mg/ml$$

   C: 1 mg/5 ml

$$\frac{1}{5} = 0.2\ mg/ml$$

   D: 1 mg/ml

   C – B – D - A

Calculating dosage

6. A patient requires a drug dose of 200 mg. The drug comes in liquid form with a strength of 5 mg/ml. How many ml does the patient need?

   Need = 200 mg

   Have 5 mg In 1 ml

$$\frac{N}{H} \times S = \frac{200\ mg}{5\ mg} \times 1\ ml = 40\ ml$$

7. A patient requires a drug dose of 500 mg. The drug comes in liquid form with a strength of 2 % w/v. How many ml does the patient need?

   Need = 500 mg
   Have 2 g in 100 ml

$$\frac{N}{H} \times S = \frac{500\ mg}{2\ g} \times 100\ ml = \frac{500\ mg}{2000\ mg} \times 100\ ml = 25\ ml$$

8. A patient requires a drug dose of 2 g. The drug comes in liquid form with a strength of 800 mg/10 ml. How many ml does the patient need?

$$\frac{2\ g}{800\ mg} \times 10\ ml = \frac{2000\ mg}{800\ mg} \times 10\ ml = 25\ ml$$

9. A patient is given a drug that comes in a strength of 0.4% w/v. The patient needs 20 mg of the drug, twice daily for 7 days. What is the total number of ml required for the treatment?

$$20\ mg \times 2 \times 7 = 280\ mg$$
$$\frac{280\ mg}{0.4\ g} \times 100\ ml = \frac{280\ mg}{400\ mg} \times 100\ ml = 70\ ml$$

10. A particular drug comes in a strength of 10 % v/v. The patient needs an active dose of 5 ml per day for 5 days, followed by 10 days of 2.5 ml per day. What is the total ml needed to complete the treatment?

$$5\ \times 5 + 2.5 \times 10 = 50\ ml$$
$$\frac{50\ ml}{10\ ml} \times 100\ ml = 500\ ml$$